SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	Slot A /2024	**Date:**	21-10-2026

Code of Conduct

I _G. Poojitha (192425337) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Poojitha

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L
= 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator
using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Q1 | Histogram Construction and Cumulative Distribution

Problem interpretation

The given grayscale image is represented by eight pixel intensities: 10, 10, 20, 20, 20, 30, 30 and 40. A histogram records how many pixels have each intensity. It does not show where pixels occur in the image; it summarizes the distribution of brightness values. The total number of pixels is N = 8, so each pixel contributes 1/8 = 0.125 to the normalized histogram and to the cumulative probability.

Histogram frequency

Count each distinct intensity directly from the list. Intensity 10 appears twice, intensity 20 appears three times, intensity 30 appears twice, and intensity 40 appears once. All other intensity levels from 0 to 255 have frequency zero. The frequencies add to 2 + 3 + 2 + 1 = 8, which verifies that every pixel has been counted.

Frequency and cumulative table

Intensity r	Frequency n(r)	Normalized frequency	Cumulative count	CDF
10	2	2/8 = 0.250	2	0.250
20	3	3/8 = 0.375	5	0.625
30	2	2/8 = 0.250	7	0.875
40	1	1/8 = 0.125	8	1.000

Cumulative distribution function

The cumulative distribution function (CDF) is the running total of normalized frequencies up to a particular intensity. For intensity 10, the cumulative count is 2 and CDF = 2/8 = 0.25. For intensity 20, the cumulative count is 2 + 3 = 5 and CDF = 5/8 = 0.625. For intensity 30, it is 7/8 = 0.875. At intensity 40, all eight pixels have been included, so CDF = 1.00. The CDF must never decrease and its final value must equal 1.

Equalization mapping

The supplied formula is s = round((L − 1) × CDF), with L = 256. Therefore L − 1 = 255. For intensity 10, s = round(255 × 0.25) = round(63.75) = 64. For intensity 20, s = round(255 ×

0.625) = round(159.375) = 159. For intensity 30, s = round(255 × 0.875) = round(223.125) = 223. For intensity 40, s = round(255 × 1) = 255. These values are the new output intensities assigned to every pixel having the corresponding original intensity.

Intensity transformation table

Original intensity	CDF	Calculation	Equalized intensity
10	0.250	round(255 × 0.250)	64
20	0.625	round(255 × 0.625)	159
30	0.875	round(255 × 0.875)	223
40	1.000	round(255 × 1.000)	255

Before-and-after comparison

Before equalization, all pixel values lie in the narrow range 10–40. The difference between the minimum and maximum is only 30 intensity levels out of the available 256, so the image has a compressed brightness range. After applying the requested mapping, the four observed levels become 64, 159, 223 and 255. The output range expands to 191 levels, from 64 to 255. This larger spread can make differences between brightness groups easier to distinguish.

Interpretation and limitations

Histogram equalization is a contrast-enhancement technique. It uses the cumulative distribution to remap pixel intensities rather than changing the spatial arrangement of pixels. In this example, pixels that were originally 10 remain equal to one another after mapping to 64; pixels originally at 20 all become 159, and so on. Equalization therefore expands separation between the four intensity groups but does not create new spatial detail. The output histogram still contains only four occupied intensity levels because the input had only four distinct values. In real images, equalization can improve visibility in low-contrast regions, but it can also amplify noise or produce an unnatural appearance when brightness distributions are already suitable.

Q2 | Averaging Filter: Neighbourhood and Calculation

Problem interpretation

The 3 × 3 neighbourhood is [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. The centre pixel is 100, which is much larger than its neighbours. This isolated high value can represent impulse noise, such as a bright pixel introduced by a sensor or transmission error. A 3 × 3 averaging filter replaces the centre with the arithmetic mean of all nine values, including the centre itself.

Mean calculation

Add all nine intensities: 10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30 = 260. The filter contains nine coefficients, each equal to 1/9, so the average is 260/9 = 28.888…. Rounded to the nearest integer, the new centre value is 29. The other eight pixels remain unchanged as directed in the question; only the centre value is replaced for this exercise.

Mean filter calculation

Neighbourhood values	Sum	Number of pixels	Mean	New centre
10, 20, 30, 20, 100, 20, 10, 20, 30	260	9	260 ÷ 9 = 28.89	29

Median calculation

A median filter sorts the nine neighbourhood values and selects the middle, fifth value. The ordered list is 10, 10, 20, 20, 20, 20, 30, 30, 100. Because there are nine values, the fifth value is the median: 20. Thus, the median-filtered centre pixel is 20.

Updated neighbourhoods

For the averaging filter, replace only 100 with 29. The resulting matrix is [[10, 20, 30], [20, 29, 20], [10, 20, 30]]. For the median filter, replace only 100 with 20. The resulting matrix is [[10, 20, 30], [20, 20, 20], [10, 20, 30]]. The median calculation is based on rank rather than the magnitude of the extreme value, which is why the value 100 does not pull the result upward as strongly as it does in the mean calculation.

Sorted values and median

Sorted position	1	2	3	4	5 (median)	6	7	8	9
Intensity	10	10	10	20	20	20	30	30	100

Filter comparison

The averaging result is 29, whereas the median result is 20. The mean is affected by every value and is pulled upward by the outlier 100. The median depends on the middle-ranked value and is resistant to one unusually large or small sample. In the given neighbourhood, 20 is also consistent with the repeated surrounding values, so the median better represents the local background intensity.

Which filter preserves details?

For salt-and-pepper or impulse noise, the median filter usually preserves edges and small structures better than an averaging filter. Averaging combines nearby values and can blur boundaries, because pixels on opposite sides of an edge contribute to the same mean. Median filtering removes isolated extremes while often retaining sharp transitions. However, neither filter is universally best: averaging can be useful for reducing approximately Gaussian random noise, while median filtering is particularly effective for isolated outliers. The choice depends on the noise model and the image details that must be retained.

Q3 | Laplacian Mask and Centre Response

Input and objective

The grayscale matrix is [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. The centre pixel has intensity 100 and its four direct neighbours each have intensity 50. The given Laplacian mask is [[0, −1, 0], [−1, 4, −1], [0, −1, 0]]. This mask responds to a difference between the centre and its immediate neighbours. A strong response indicates a rapid local intensity transition.

Convolution at the centre

Multiply each image value by the mask value at the matching position and add the products. The four corner positions have mask coefficient 0 and contribute nothing. The centre contributes 4 × 100 = 400. Each of the four direct neighbours contributes (−1) × 50 = −50. The total response is 400 − 50 − 50 − 50 − 50 = 200. Therefore, the Laplacian response at the centre is +200.

Sharpening calculation summary

Centre contribution	Four neighbour contributions	Laplacian response	Original centre	Sharpened centre
4 × 100 = 400	4 × (−50) = −200	400 − 200 = 200	100	100 + 200 = 300

Centre pixel after sharpening

The question specifies that the response is added to the original centre intensity. The original centre is 100 and the calculated Laplacian response is 200. Therefore, the sharpened value is 100

+ 200 = 300. This is the direct mathematical result requested by the question. If the result is stored in a standard 8-bit grayscale image, valid values are limited to 0–255, so 300 would normally be clipped to 255 or rescaled. The calculation should still be reported as 300 before any display-range handling.

Sharpening interpretation

Sharpening increases local contrast around intensity changes so that boundaries and fine structures appear clearer. Here, the centre is brighter than all four direct neighbours. The mask produces a positive response because the centre is substantially higher than its neighbours. Adding this response increases the centre value further, emphasizing the local bright feature.

Role of second-order intensity changes

The Laplacian is a second-order spatial derivative. Unlike a simple smoothing filter, it emphasizes locations where intensity changes rapidly. Uniform regions tend to produce a small response, while edges, thin lines, corners and small bright or dark features can produce larger positive or negative responses. The supplied mask uses the centre coefficient 4 and four negative neighbour coefficients, making the response equal to four times the centre minus the sum of the four direct neighbours.

Practical considerations

Laplacian sharpening can improve visual clarity in medical images, industrial inspection, document enhancement and machine-vision preprocessing. Nevertheless, derivatives also amplify high-frequency noise. A noisy image may therefore require mild smoothing before sharpening. The sign convention matters: this question explicitly asks to add the response, so the calculated output is 300. Other implementations may use a mask with the opposite sign or subtract the Laplacian response; those conventions must be matched to the chosen mask. For an 8-bit image, clipping, normalization or a wider intermediate data type is needed to avoid overflow and preserve a meaningful result.

Q4 | Global Thresholding Rule and Pixel Classification

Problem interpretation

The grayscale image is [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Global thresholding converts each grayscale pixel into one of two labels using a single threshold for the entire image. The specified threshold is T = 100. According to the rule, a pixel is assigned 1 when its intensity is greater than or equal to 100; otherwise it is assigned 0. In this answer, 1 denotes foreground and 0 denotes background.

Classification method

Examine each pixel independently and compare it with 100. Values 40, 60, 80 and 30 are below the threshold and become 0. Values 120, 150, 200, 100 and 180 are equal to or above the threshold and become 1. Importantly, the value 100 itself is foreground because the stated rule uses greater than or equal to the threshold, not strictly greater than it.

Binary classification table

Original value	40	60	120	80	150	200	30	100	180
Binary output	0	0	1	0	1	1	0	1	1

Resulting binary matrix

Applying the rule row by row gives the first row [0, 0, 1], because 40 and 60 are below 100 while 120 is above it. The second row becomes [0, 1, 1], because 80 is below the threshold and 150 and 200 meet it. The third row becomes [0, 1, 1], because 30 is below the threshold, while 100 and 180 meet the threshold.

Foreground and background totals

The final binary image is [[0, 0, 1], [0, 1, 1], [0, 1, 1]]. Count the ones: there is one in row 1, two in row 2 and two in row 3, giving 1 + 2 + 2 = 5 foreground pixels. Count the zeros: there are two in row 1, one in row 2 and one in row 3, giving 2 + 1 + 1 = 4 background pixels. The check is 5 + 4 = 9, matching the nine pixels in the original 3 × 3 image.

Sobel Gx row-wise calculation

Row	Weighted calculation	Contribution
Top	$-10 + 0 + 10$	0
Middle	$-20 + 0 + 180$	160
Bottom	$-10 + 0 + 90$	80
Total	$0 + 160 + 80$	$G_x = 240$

How the threshold separates regions

Thresholding is a basic image-segmentation technique because it groups pixels according to brightness. When an object is brighter than its background, a threshold can isolate the object as foreground. The binary image is easier to process than the original grayscale matrix and can support later operations such as counting objects, measuring area, finding connected components or locating approximate boundaries. A global threshold is computationally simple and produces a predictable binary result.

Limitation and possible improvement

One important limitation is sensitivity to illumination variation. If the same object is bright in one part of an image and shaded in another, a single global threshold may classify some object pixels as background and some background pixels as foreground. The method also performs poorly when object and background intensities overlap or when the image contains significant noise. Possible alternatives include adaptive or local thresholding, which calculates a threshold from a neighbourhood, and preprocessing such as denoising or contrast adjustment. The best method depends on lighting, contrast and the segmentation objective; the current answer uses $T = 100$ exactly as specified.

Sobel Gy row-wise calculation

Row	Weighted calculation	Contribution
Top	$-10 - 20 - 10$	-40
Middle	0	0
Bottom	$10 + 100 + 90$	200
Total	$-40 + 0 + 200$	$G_y = 160$

Q5 | Sobel Horizontal Gradient Component

Input and operator

The grayscale matrix is [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. The Sobel operator estimates the first-order intensity gradient using two 3 × 3 masks. Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] measures intensity change in the horizontal x direction. Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] measures change in the vertical y direction. The two components together describe the strength and direction of local change.

Calculation of Gx

Multiply each image pixel by its corresponding Gx coefficient and sum. The top row contributes (−1 × 10) + (0 × 10) + (1 × 10) = 0. The middle row contributes (−2 × 10) + (0 × 50) + (2 × 90) = −20 + 180 = 160. The bottom row contributes (−1 × 10) + (0 × 50) + (1 × 90) = 80. Adding the row contributions gives Gx = 0 + 160 + 80 = 240.

Calculation of Gy

Apply the Gy mask to the same 3 × 3 neighbourhood. The top row contributes (−1 × 10) + (−2 × 10) + (−1 × 10) = −40. The middle row contributes zero because all three mask coefficients are zero. The bottom row contributes (1 × 10) + (2 × 50) + (1 × 90) = 10 + 100 + 90 = 200. The total vertical component is Gy = −40 + 0 + 200 = 160.

Gradient magnitude

The gradient magnitude combines both directional components and is calculated using $G = \sqrt{(Gx^2 + Gy^2)}$. Substituting the results gives $G = \sqrt{(240^2 + 160^2)} = \sqrt{(57{,}600 + 25{,}600)} = \sqrt{83{,}200} \approx 288.44$. This positive magnitude represents the strength of the local intensity change. A larger gradient magnitude generally indicates a stronger candidate edge, although the practical detection threshold depends on image scale, noise and the application.

Direction interpretation

The horizontal component is Gx = 240 and the vertical component is Gy = 160. Since |Gx| is greater than |Gy|, the intensity changes more strongly in the x direction. The gradient angle, measured from the positive x axis, is $\theta = \text{atan2}(Gy, Gx) = \text{atan2}(160, 240) \approx 33.7°$. The gradient points in the direction of greatest intensity increase. The visible edge itself is perpendicular to

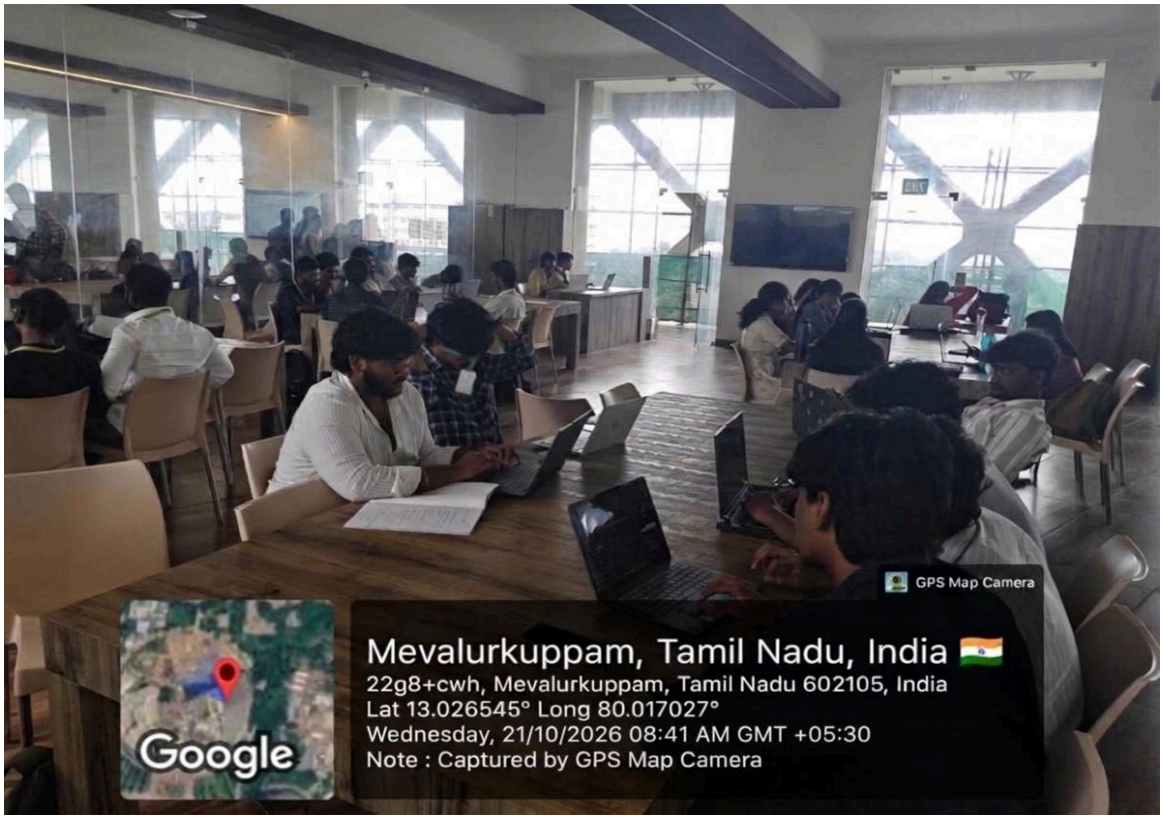
the gradient, so its orientation is approximately 33.7° + 90° = 123.7° (equivalently, −56.3°). Thus the edge is more vertical than horizontal.

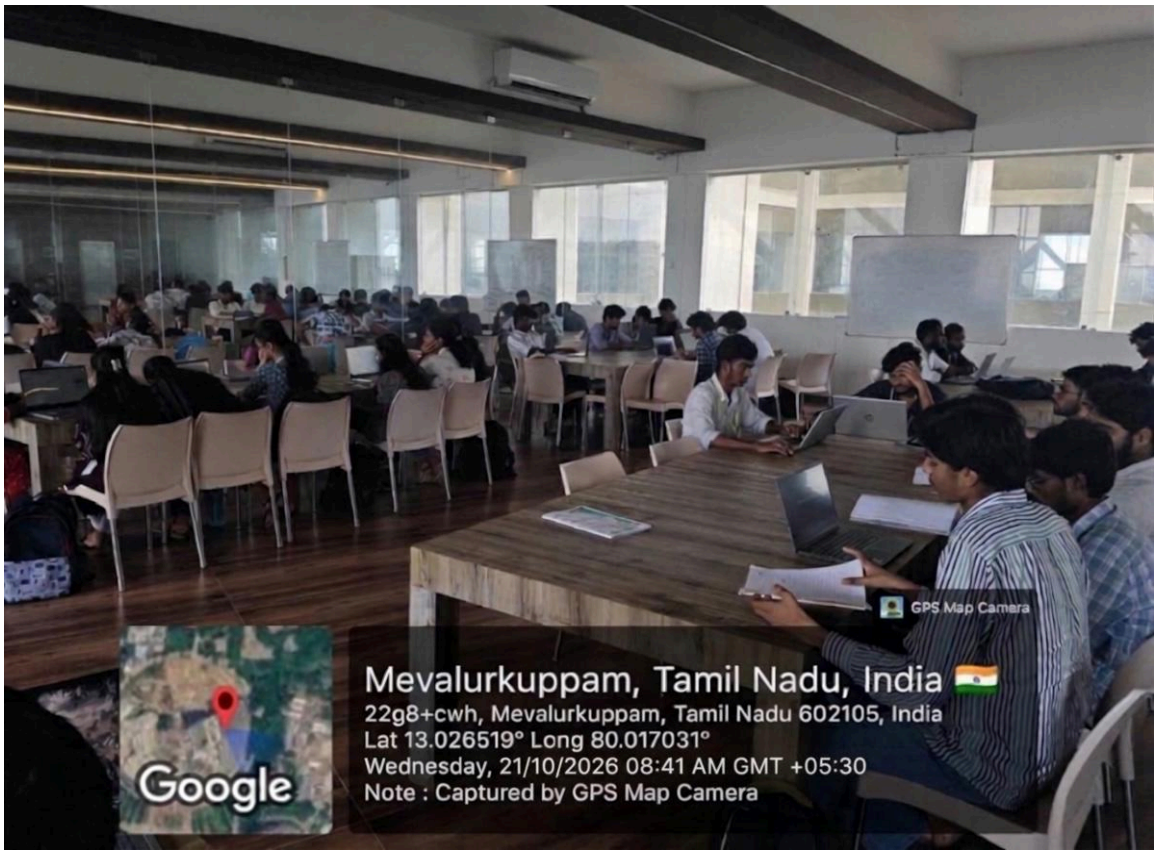
Use in boundary detection and segmentation

Sobel gradients help identify object boundaries because pixels near a boundary often have a strong intensity transition. A vision pipeline can compute Gx and Gy, calculate gradient magnitude, suppress weak responses with a threshold and connect neighbouring edge pixels into contours. The resulting boundaries can support segmentation by separating regions, estimating object shape, measuring dimensions or locating parts for inspection. Gradient direction can also guide contour tracing and edge linking. However, Sobel responses can be affected by noise and texture; smoothing, non-maximum suppression, threshold selection or more advanced methods may be needed for reliable boundaries.

Calculation summary

Quantity	Result	Meaning
Gx	240	Horizontal intensity gradient
Gy	160	Vertical intensity gradient
Magnitude	$\sqrt{83{,}200} \approx 288.44$	Strength of local change
Gradient angle	≈ 33.7°	Direction of greatest increase
Edge orientation	≈ 123.7°	Perpendicular to gradient





This activity has been discussed in class session and completed the assessment .

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
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